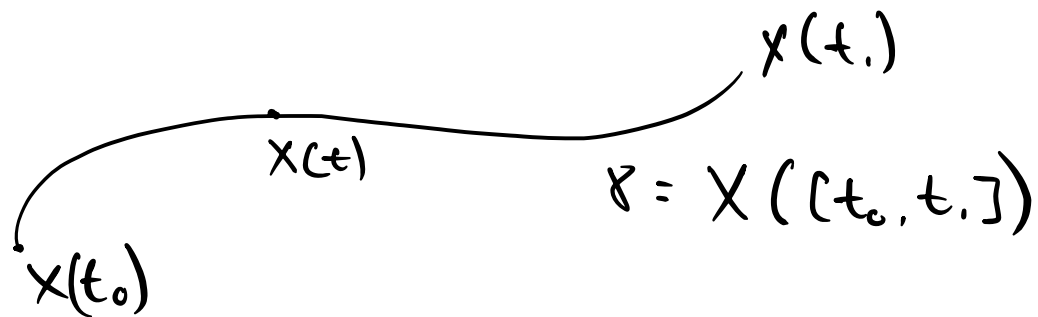


String Theory Basics

- Lagrange mechanics for point particles
- Adapt this to strings
- Compactification & T-duality

For a curve, γ , sitting in $\mathbb{R}^n$,
can parameterise by $X^i: [t_0, t_1] \rightarrow \mathbb{R}^n$

$$\dot{X}^i(t) = (\dot{x}^1(t), \dots, \dot{x}^n(t))$$



It's arc-length param. given by

$$ds = |\dot{X}(t)| dt$$

$$l = \int_{t_0}^{t_1} |\dot{X}(t)| dt$$

$|\cdot|$ is just the norm in Euclidean metric

(As a matrix $|u| = \sqrt{u^T \cdot g \cdot u}$ $g = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$)

But can consider other metrics,

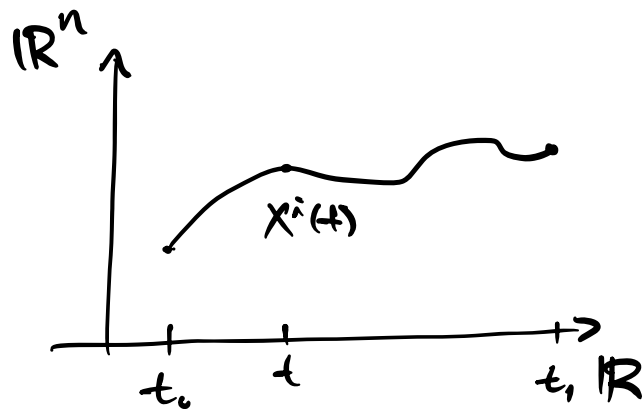
We are interested in Minkowski metric in $\mathbb{R} \times \mathbb{R}^n$

$\eta = \begin{pmatrix} \overset{\text{speed of light}}{\downarrow} c^2 & & 0 \\ & -1 & \\ 0 & & \ddots \\ & & & -1 \end{pmatrix}$

(Lorentz transformations of special relativity leave η invariant)

Point Particle in Minkowski Space

A particle will trace out a world-line in $\mathbb{R} \times \mathbb{R}^n$ as it moves through time (and space).



Now parameterised by $X^i(t) = (X^0(t), X^1(t), \dots, X^n(t))$

Arc-length param. is

$$ds = \sqrt{\eta \left(\frac{\partial x}{\partial t}, \frac{\partial x}{\partial t} \right)} dt = \sqrt{\sum_{ij} \eta_{ij} \frac{\partial x^i}{\partial t} \frac{\partial x^j}{\partial t}} dt$$

Hence the length of the world-line is

$$l = \int_{t_0}^{t_1} \sqrt{\eta \left(\frac{\partial x}{\partial t}, \frac{\partial x}{\partial t} \right)} dt$$

Multiplying the length by some constants for unit reasons gives us the action

$$S = - \overset{\text{particle mass}}{\downarrow} mc \int ds = - mc \int_{t_0}^{t_1} \sqrt{\sum_{ij} \eta_{ij} \frac{\partial x^i}{\partial t} \frac{\partial x^j}{\partial t}} dt$$

Everything is derived from the action

In general, an action is of the form

$$S = \int L dt$$

↖ Lagrangian

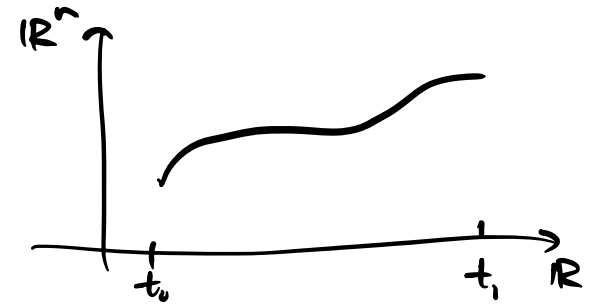
Almost all field theories stem from an action.

Can be thought of as "energy", and hence anything physical should "minimise" the action, $\delta S = 0$.

↑
or just extremise

Point Particle in Minkowski Space

In our case, assuming $x^0(t) = t$



$$L = -mc \sqrt{\sum_{i,j} \eta_{ij} \frac{\partial x^i}{\partial t} \frac{\partial x^j}{\partial t}} = -mc \sqrt{c^2 - \sum_{\mu=1}^n \left(\frac{\partial x^\mu}{\partial t} \right)^2}$$

$$= -mc^2 \sqrt{1 - \frac{\vec{v}^2}{c^2}}$$

← speed of the particle in space squared.

Harder to work with this Lagrangian.

Lets work in the "classical limit", i.e take $\vec{v} \ll c$.

Then
$$L_{cl} = \frac{1}{2}mv^2 - mc^2$$

(Note its form:
$$L = [\text{kinetic energy}] - [\text{potential energy}]$$

use this in a minute!)

Now computing $\delta S_{\text{cl}} = 0$ gives the equations of motion as

$$\frac{\partial \vec{v}}{\partial t} = 0 \quad \text{travels in a straight line}$$

Also need: Conjugate momentum

$$\pi^i = \frac{\partial L}{\partial \dot{x}^i} \leftarrow \text{time derivative of param.}$$

free non-relativistic point particle: $\pi^i = \underbrace{m v^i}_{\text{usual momentum!}}$

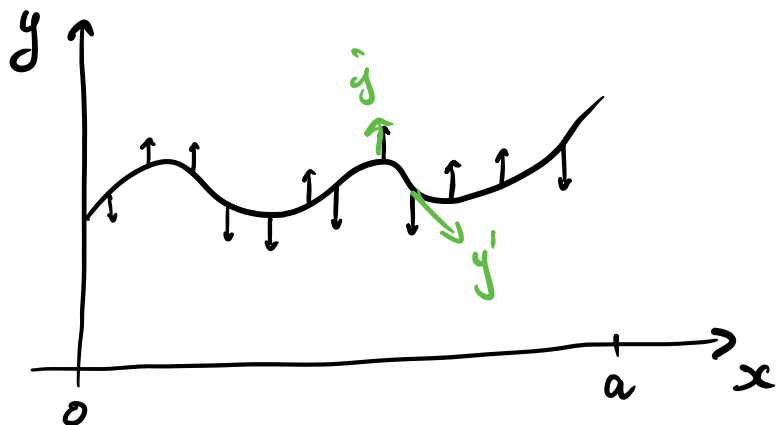
free relativistic point particle: $\pi^i = \frac{m v^i}{\underbrace{\sqrt{1 - \frac{v^2}{c^2}}}_{\text{usual relativistic momentum!}}}$

Warning: not true in general.

Strings

Non-relativistic string

Consider a string in 2-dimensions, moving in only one direction



Graph of a function $y = f(x)$
which also moves over time,
 $y(t)$

At a point x , the string has velocity y

— " — , the string is "stretched" by y'

$\frac{\partial y}{\partial t}$



$\frac{\partial y}{\partial x}$

Thus, total kinetic energy is $K = \int_0^a \frac{1}{2} \mu_0 (\dot{y})^2 dx$ constant mass density

total potential energy $V = \int_0^a \frac{1}{2} T_0 (y')^2 dx$ constant string tension.

Then, letting $L = K - V$, gives us an action, S .

Can work out equations of motion, $\delta S = 0$

$$y'' - \frac{\mu_0}{T_0} \ddot{y} = 0 \quad \text{wave equation}$$

There are now two conjugate momenta

$$\pi_t = \frac{\partial L}{\partial \dot{y}} = \mu_0 \dot{y}, \quad \pi_x = \frac{\partial L}{\partial y'} = -T_0 y' \Rightarrow \boxed{\frac{\partial \pi^t}{\partial t} + \frac{\partial \pi^x}{\partial x} = 0}$$

E.O.M are

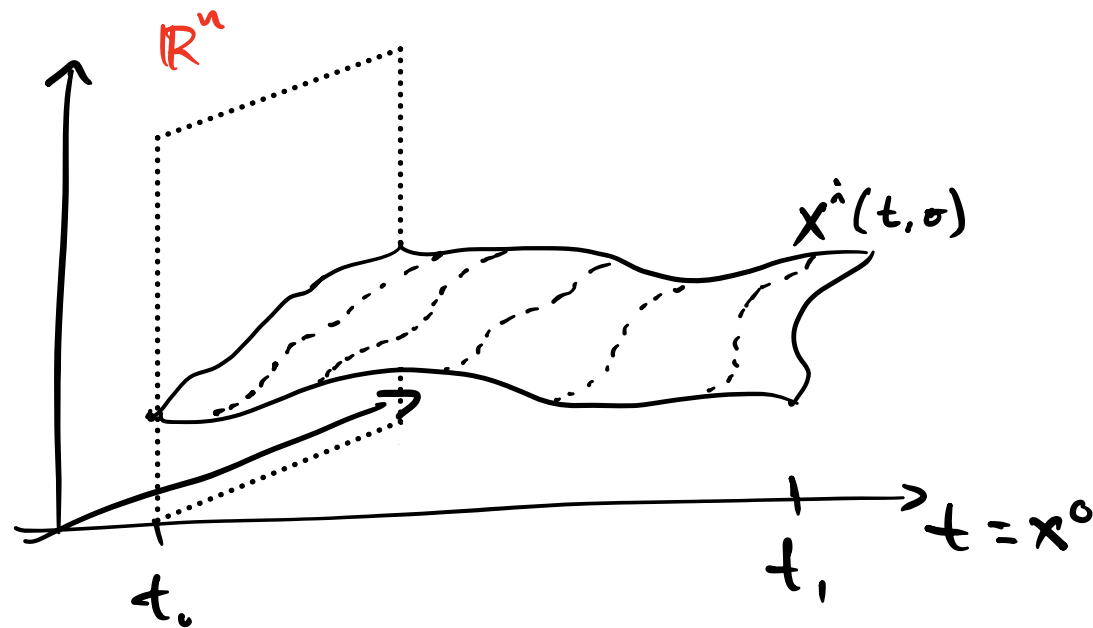
Also, total momentum of the string is given by

$$p = \int_0^a \pi^t dx$$

Relativistic string

Working in $\mathbb{R} \times \mathbb{R}^n$ again, our string traces out a world-sheet, param. by t & σ as $X^i(t, \sigma)$, $i=0, \dots, n$

$\uparrow$
space coord.



Minkowski

$$\eta = \begin{pmatrix} c^2 & & 0 \\ & -1 & \\ 0 & & \ddots \\ & & & -1 \end{pmatrix}$$

The action will now involve the area of the world-sheet
(c.f. length for world-line of point particle).

Denote $g = \begin{pmatrix} \eta(\underline{\dot{x}}, \underline{\dot{x}}) & \eta(\underline{\dot{x}}, \underline{\dot{x}}') \\ \eta(\underline{\dot{x}}', \underline{\dot{x}}) & \eta(\underline{\dot{x}}', \underline{\dot{x}}') \end{pmatrix}$ (denote $\eta(\dot{x}, \dot{x}) = \dot{x}^2$
 $\eta(\dot{x}, \dot{x}') = \dot{x} \cdot \dot{x}'$
etc.)

The area element is then

$$dA = \sqrt{-\det g} dt d\sigma, \quad \text{Area} = \int dA$$

Hence define the action

$$S = -\frac{T_0}{c} \int dA = -\frac{T_0}{c} \int_{t_0}^{t_1} \int_{\sigma_0}^{\sigma_1} \underbrace{\sqrt{(\dot{x} \cdot \dot{x}')^2 - (\dot{x})^2 (\dot{x}')^2}}_{=L} d\sigma dt$$

conjugate momenta:

$$\pi_t^i = \frac{\partial L}{\partial \dot{x}^i}$$

$$\pi_\sigma^i = \frac{\partial L}{\partial \dot{x}'^i}$$

With this, equations of motion are

$$\frac{\partial \pi_t^i}{\partial t} + \frac{\partial \pi_\sigma^i}{\partial \sigma} = 0$$

same as for
non-relativistic string!

However, we do not have the wave equation.

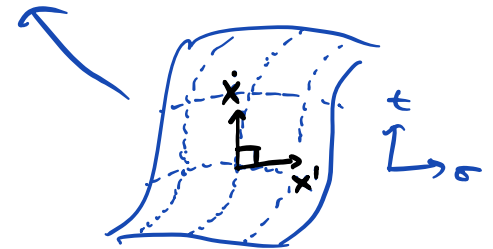
Canonical momenta are much more complicated.

Not all hope lost! The action is parameterisation invariant.

more like constraints
coming from gauge condition

If we pick param. such that $\dot{X} \cdot X' = 0$ and

$$\dot{X}^2 + X'^2 = 0$$



↑
says that contraction from
the string change its velocity.

Then conjugate momenta simplify to

$$\pi_t^i = \frac{\partial \mathcal{L}}{\partial \dot{X}^i} = \frac{1}{2\pi\alpha'} \dot{X}^i$$

$$\pi_\sigma^i = \frac{\partial \mathcal{L}}{\partial X'^i} = -\frac{1}{2\pi\alpha'} X'^i$$

$$\alpha' = \frac{1}{2\pi T_0 c}$$

"length of string squared"

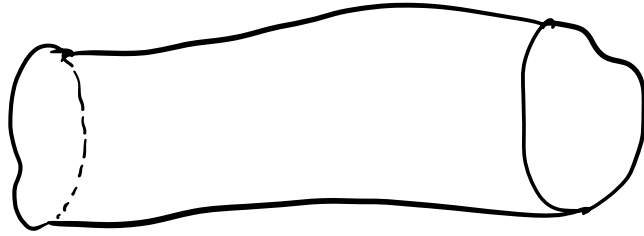
This turns the equations of motion into our n-dimensional wave equation

$$\ddot{X}^i - X''^i = 0$$

We will now turn to solving this.

The closed string.

Assume we have a closed string, i.e. $X^i(t, 0) = X^i(t, 2\pi)$



world-sheet
becomes
world-tube.

Hence we get 2π -periodicity.

Wave equation has a left and right moving solution

$$X^i(t, \sigma) = X_L^i(t + \sigma) + X_R^i(t - \sigma)$$

$\sigma^\pm = t \pm \sigma$
"light cone
coords".

which have solutions

$$X_L^i(\sigma^+) = \frac{1}{2} x_0^{Li} + \sqrt{\frac{\alpha'}{2}} p_0^{Li} \sigma^+ + \text{oscillatory terms}$$

$$X_R^i(\sigma^-) = \frac{1}{2} x_0^{Ri} + \sqrt{\frac{\alpha'}{2}} p_0^{Ri} \sigma^- + \text{oscillatory terms}$$

centre of mass movement

Periodicity dictates $p_0^{R,i} = p_0^{L,i} =: p_0^i$ "level matching condition"

Then

$$X^i(t, \sigma) = x_0^i + \sqrt{2\alpha'} p_0^i t + \text{oscillatory terms}$$

Conjugate momentum is $\pi_t^i = \frac{1}{2\pi\alpha'} \dot{X}^i + \text{osc.}$

hence total momentum is

$$P^i = \int_0^{2\pi} \frac{1}{\pi\sqrt{2\alpha'}} p_0^i d\sigma = \sqrt{\frac{2}{\alpha'}} p_0^i$$

T-Duality

(For reasons) string theory must take place in 26 dimensions. We only perceive 4, so dimensions must be hidden somewhere. Dealt with via compactification.

Consider "1D" space time with an extra compactified dimension, $\mathbb{R} \times \mathbb{R} \times S^1$

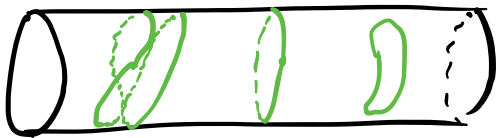
$\uparrow$ time $\uparrow$ space $\uparrow$ compact.

space :



closed strings have many more varieties now

$\pi_1(S^1) = \mathbb{Z}$, associate an integer m to the string.



winding number is $w = \frac{m R}{\alpha'}$

$\downarrow$ S^1 radius.

only considering last coord, suppressing index

Periodicity becomes $X(t, \sigma + 2\pi) = X(t, \sigma) + 2\pi\alpha'\omega$
and level matching becomes

$$\text{and also have } \left. \begin{aligned} \frac{1}{\sqrt{2\alpha'}} (p_0^L - p_0^R) &= \omega \\ \frac{1}{\sqrt{2\alpha'}} (p_0^L + p_0^R) &= p \end{aligned} \right\} \Rightarrow p \pm \omega = \sqrt{\frac{2}{\alpha'}} p_0^{\frac{L}{R}}$$

Let q_0 be s.t. $x_0 + q_0 = x_0^L$, $x_0 - q_0 = x_0^R$. Then

$$X_L(\sigma^+) = \frac{1}{2} (x_0 + q_0) + \frac{\alpha'}{2} (p + \omega) \sigma^+ + \text{osc.}$$

$$X_R(\sigma^-) = \frac{1}{2} (x_0 - q_0) + \frac{\alpha'}{2} (p - \omega) \sigma^- + \text{osc.}$$

We have $X = X_L + X_R$, define $\tilde{X} = X_L - X_R$

$$X(t, \sigma) = x_0 + \alpha' p t + \alpha' \omega \sigma + \text{osc}$$

$$\tilde{X}(t, \sigma) = q_0 + \alpha' \omega t + \alpha' p \sigma + \text{osc}$$

conjugate momentum of new coords 'i' $\tilde{\pi}_t = \frac{1}{2\pi\alpha'} \dot{\tilde{X}}$

$$\Rightarrow \tilde{p} = \int_0^{2\pi} \tilde{\pi}_t d\sigma = \omega$$

and also $\tilde{X}(t, \sigma + 2\pi) - \tilde{X}(t, \sigma) = 2\pi\alpha' p$ winding no. for $\tilde{X}$.

So new coordinates interchange ω & p !

Quantum mechanics (briefly)

Translation operator on a vector $|x\rangle$, $T(a)|x\rangle = |x+a\rangle$
may be written $T(a) = e^{-iap}$.

Thus $T(2\pi R)|x\rangle = |x+2\pi R\rangle = |x\rangle \Rightarrow 2\pi R p = 2\pi n \quad n \in \mathbb{Z}$
 $\Rightarrow p = \frac{n}{R}$ quantized!

Fact: $M^2 = p^2 + \omega^2 + \frac{2}{\alpha'} (\underbrace{N + \tilde{N}}_{\text{number operators}} - 2)$

using $p = \frac{n}{R}$, $\omega = \frac{m R}{\alpha'}$

$$M^2 = \frac{n^2}{R^2} + \frac{m^2 R^2}{\alpha'^2} + \frac{2}{\alpha'} (N + \tilde{N} - 2)$$

If we change to coords $(X, \pi) \rightarrow (\tilde{X}, \tilde{\pi})$, i.e. $p \leftrightarrow \omega$
whilst changing the radius $R \rightarrow \tilde{R} = \frac{\alpha'}{R}$, nothing changes!

This is T-duality

Can show that the level matching condition is $N - \tilde{N} = pw$.
 Can express this and the mass condition as

$$M^2 = Z \mathcal{H} Z^t + (N + \tilde{N} - 2)$$

$$N - \tilde{N} = Z \mathcal{J} Z^t$$

$$\mathcal{H} = \begin{pmatrix} \frac{R}{\alpha'} & 0 \\ 0 & \frac{1}{R} \end{pmatrix}$$

$$\mathcal{J} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Z = \begin{pmatrix} m & n \end{pmatrix}$$

Can view T-duality as an element $h \in O(1,1, \mathbb{Z})$.

There are only two elements of this group, I , $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

↑
 isometries of $(1,1)$ signature metric i.e. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

actions: $h^t \mathcal{J} h = \mathcal{J}$, $\begin{pmatrix} m & n \end{pmatrix} h = \begin{pmatrix} n & m \end{pmatrix}$, $h^t \begin{pmatrix} \frac{R}{\alpha'} & 0 \\ 0 & \frac{1}{R} \end{pmatrix} h = \begin{pmatrix} \frac{1}{R} & 0 \\ 0 & \frac{R}{\alpha'} \end{pmatrix}$

$n \leftrightarrow m$ $R \leftrightarrow \frac{\alpha'}{R}$

General T-duality

Now consider compactification T^d . T^d has tensor G_{ij} & B_{ij} describing its geometry (e.g. S^1 ; $G=R$, $B=0$).

Can write

$$\mathcal{H} = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}$$

$$M^2 = Z \mathcal{H} Z^t + (N + \tilde{N} - 2)$$

$$N - \tilde{N} = Z J Z^t$$

$$J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

$$Z = \begin{pmatrix} m^i & n^i \end{pmatrix}$$

Hence see T-duality as an $h \in O(d, d, \mathbb{Z})$

$O(d,d)$ generated by following:

B-field shift by Θ : $h_{\Theta} = \begin{pmatrix} I & \Theta \\ 0 & I \end{pmatrix} \rightsquigarrow B \mapsto B + \Theta$

skew

Basis change by A : $h_A = \begin{pmatrix} A & 0 \\ 0 & (A^T)^{-1} \end{pmatrix} \rightsquigarrow$ Large diffeomorphisms of T^d

$G_2(d, \mathbb{Z})$

Factorized duality $\mathcal{D}_i$: $h_{\mathcal{D}_i} = \begin{pmatrix} I - \mathcal{D}_i & \mathcal{D}_i \\ \mathcal{D}_i & I - \mathcal{D}_i \end{pmatrix}$

$\mathcal{D}_i = \begin{pmatrix} 0 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \ddots & \\ & & & & 0 \end{pmatrix} \rightsquigarrow$ generalisation of T-duality $R \rightarrow \alpha'/R$.

Thank you !